Abstract

This report presents a comprehensive experimental study on financial sentiment analysis using modern deep learning techniques and fine-tuned large language models. The research addresses the challenge of classifying sentiment in financial text, where domain-specific vocabulary and subtle expression patterns make accurate prediction difficult for traditional NLP systems. We adopt a well-structured pipeline that begins with dataset preparation and preprocessing, continues through parameter-efficient fine-tuning, and concludes with rigorous evaluation on test samples.

The project fine-tunes three advanced LLMs—Llama-3, Qwen-2.5, and DeepSeek-R1 Distill—using Low-Rank Adaptation (LoRA) to reduce training overhead while preserving model capacity. Input financial statements are converted from raw CSV into JSONL format and normalized to support a three-way sentiment classification task (positive, neutral, negative). The training workflow leverages GPU acceleration and careful hyperparameter tuning to maximize model stability and generalization.

Experimental evaluation uses standard classification metrics, including accuracy, precision, recall, and F1-score, to compare model behavior across different sentiment categories. In addition, the system architecture incorporates Dask for distributed preprocessing and batch inference, demonstrating a scalable approach to handling larger financial datasets. This Big Data integration enables the pipeline to process text efficiently and supports future expansion to real-world financial analytics workloads.

The results show that fine-tuned LLMs can effectively distinguish nuanced sentiment in financial narratives, outperforming baseline approaches and delivering strong classification performance. Llama-3 performs particularly well in overall accuracy, Qwen-2.5 balances efficiency and robustness, and DeepSeek-R1 Distill offers improved interpretability through reasoning traces. The combined findings highlight the value of LLM fine-tuning in financial sentiment tasks and the importance of distributed processing for scalability.

Overall, this research contributes practical methodology for applying deep learning and large language models to financial sentiment analysis. It demonstrates how modern NLP techniques and distributed computing can work together to produce accurate, scalable, and explainable sentiment classification solutions for finance-related applications.

Keywords: Sentiment Analysis, Financial Analysis, Deep Learning, Large Language Models (LLMs), Fine-tuning, LoRA, Dask, Distributed Computing, Big Data